Software Requirements Specification (SRS) Template

Project: Traffic Management System (Simulation)
Version: 1.0
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Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

The purpose of the Traffic Management System (Simulation) is to simulate and manage traffic flow at a road intersection using queue-based data structures. The system allows vehicles to be added to multiple lanes, processed according to their priority, and monitored through traffic density, queue status, traffic signals, and statistical information. The system is developed in C and is intended to demonstrate the application of data structures and software engineering concepts in traffic management.

1.2 Scope

The Traffic Management System (Simulation) provides a simple simulation of traffic flow through multiple lanes. The system allows users to add vehicles to lanes, process vehicles from queues, give priority to emergency vehicles, manage traffic signals, monitor queue sizes, calculate traffic density, and view traffic statistics. The system supports up to four traffic lanes and maintains separate queues for each lane. It also provides input validation, handling of empty and full queues, and an option to reset the simulation. The system is implemented in C using queue-based data structures. The project is intended as a simulation and does not directly control real-world traffic signals, vehicles, or road infrastructure.

1.3 Audience

The intended audience for this document includes the project team, faculty members, evaluators, and users interested in understanding the Traffic Management System (Simulation). It also serves as a reference for developers and testers involved in implementing, testing, and evaluating the system.

1.4 Definitions

Term	Definition
TMS	Traffic Management System
FR	Functional Requirement
NFR	Non-Functional Requirement
UML	Unified Modeling Language
RTM	Requirements Traceability Matrix
CI/CD	Continuous Integration / Continuous Delivery
Queue	A data structure used to maintain vehicles waiting in a lane.
Lane	A separate queue representing a traffic lane in the simulation.
Traffic Density	A classification of traffic based on the number of vehicles waiting in a lane.
Emergency Vehicle	A vehicle given higher processing priority than normal vehicles.

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2. Overall description

2.1 Product perspective

The Traffic Management System (Simulation) is a standalone console-based application developed in C. The system simulates traffic flow through multiple lanes using queue-based data structures. Each lane maintains its own queue of vehicles, allowing vehicles to be added and processed while maintaining their priority. The system provides features for vehicle management, multiple-lane traffic handling, traffic signal status, emergency vehicle priority, traffic density monitoring, queue monitoring, statistics, simulation reset, and input validation. The system does not depend on external traffic infrastructure or real-world traffic-control systems. It operates as a software simulation and provides output through a command-line interface.

2.2 Major product functions (detailed)

The Traffic Management System provides the following major functions:

1. **Vehicle Addition:** Allows the user to add vehicles to selected traffic lanes.
2. **Vehicle Processing:** Processes vehicles from the lane queues according to their priority.
3. **Multiple Lane Management:** Maintains separate queues for multiple traffic lanes.

4. **Traffic Signal Management:** Displays and manages the traffic signal status.
5. **Emergency Vehicle Priority:** Gives priority to emergency vehicles waiting in a lane.
6. **Traffic Density Monitoring:** Determines the traffic density of each lane based on the number of waiting vehicles.
7. **Queue Monitoring:** Displays the current vehicles and queue status of the lanes.
8. **Statistics:** Maintains and displays the total number of vehicles added and processed.
9. **Simulation Reset:** Allows the user to reset the simulation and clear the current traffic data.
10. **Input Validation:** Validates user input and handles invalid selections and queue conditions.

2.3 User roles and characteristics (expanded)

The Traffic Management System has a single primary user role:
System User: The system user interacts with the console-based application to manage and monitor the traffic simulation. The user can add vehicles to lanes, process vehicles, view lane queues, manage traffic signals, monitor traffic density, view statistics, reset the simulation, and exit the system.
The system does not require separate administrator or operator accounts because it is designed as a standalone simulation.

2.4 Operating environment

The Traffic Management System operates as a standalone console-based C application on a standard computer. The system requires a C compiler such as GCC and a command-line environment for execution. It does not require specialized hardware, network connectivity, a database, or external services.

2.5 Constraints

The system is implemented using the C programming language and uses queue-based data structures for traffic management. The simulation supports a maximum of four traffic lanes and a limited queue capacity per lane. The system is designed for simulation purposes and does not interface with real-world traffic signals, sensors, vehicles, or road infrastructure. User interaction is limited to the command-line interface.

3. External interface requirements

3.1 User interfaces

The Traffic Management System uses a command-line interface (CLI). The main menu provides options for adding vehicles, processing vehicles, displaying queues, managing multiple lanes, viewing traffic signal status, monitoring traffic density, viewing statistics, resetting the simulation, and exiting the system.

The system displays clear prompts and messages to guide the user during interaction. Invalid inputs and queue conditions are handled with appropriate messages.

3.2 Hardware interfaces

The system does not require any specialized hardware. It runs on a standard computer capable of compiling and executing a C program.

3.3 Software interfaces

The system is developed using the C programming language and can be compiled using a standard C compiler such as GCC. The application runs as a standalone console program and does not require a database or external software service.

3.4 Communications

The system does not require network communication or external APIs. All interaction takes place locally through the command-line interface.

<< Make sure overall there are at least 15 FRs for overall project, 5 NFRs, 2 security objectives and 5 Security requirements>>

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow TMS-F-###.

4.1 Authentication

Description: Authenticate the cardholder using EMV card and PIN; support chip & contactless where applicable.

Req ID	**Requirement (shall...)**	**Type**	**Priority**	**Source/Stakeholder**	**Acceptance criteria / Test case ref**	**Comments / Dependencies**
TMS-F-001	The system shall allow the user to add a vehicle with a vehicle ID to a selected lane queue.	Functional	High	System User	AC-TMS-F-001: Valid vehicle ID and lane selection add the vehicle to the selected queue. Test: TC-TMS-001	Requires valid lane and available queue capacity.
TMS-F-002	The system shall process and remove	Functional	High	System User	AC-TMS-F-002: The front vehicle is	Depends on a non-empty

	the vehicle at the front of the selected lane queue.				removed and reported as processed; an empty queue is handled correctly. Test: TC-TMS-002	selected queue.
TMS-F-003	The system shall display the vehicles currently waiting in a selected lane, including vehicle IDs and priority status.	Functional	Medium	System User	AC-TMS-F-003: Queue contents and priority status are displayed correctly; an empty queue produces an appropriate message. Test: TC-TMS-003	Depends on selected lane and current queue contents.

4.2 Multiple Lane Management

Description: Maintain independent queues for the supported traffic lanes and validate lane selection.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMS-F-004	The system shall maintain a separate queue for each supported traffic lane.	Functional	High	System User	AC-TMS-F-004: A vehicle added to one lane remains in that lane and does not appear in another lane. Test: TC-TMS-004	Supports up to four lanes.
TMS-F-005	The system shall allow the user to select a valid lane before performing lane-specific operations.	Functional	High	System User	AC-TMS-F-005: Valid lane selections are accepted and invalid lane selections are rejected with an error message. Test: TC-TMS-005	Lane number must be within the supported range.
TMS-F-006	The system shall detect and report empty and full lane queues without corrupting queue data.	Functional	High	System User	AC-TMS-F-006: Processing an empty queue and adding to a full queue produce appropriate messages without invalid memory access. Test: TC-TMS-006	Depends on finite queue capacity.

4.3 Traffic Signal and Priority

Description: Provide traffic signal status and priority handling for emergency vehicles.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments /

						Dependencies
TMS-F-007	The system shall allow the user to view and set the traffic signal status as Red or Green.	Functional	High	System User	AC-TMS-F-007: The selected signal status is displayed correctly as Red or Green. Test: TC-TMS-007	Signal status is managed within the simulation.
TMS-F-008	The system shall give emergency vehicles priority over normal waiting vehicles in a lane queue.	Functional	High	System User	AC-TMS-F-008: An emergency vehicle added while normal vehicles are waiting is positioned ahead of normal waiting vehicles. Test: TC-TMS-008	Uses priority insertion in the queue.
TMS-F-009	The system shall display whether a waiting vehicle is an emergency vehicle when queue information is shown.	Functional	Medium	System User	AC-TMS-F-009: Emergency and normal vehicles are distinguishable in displayed queue information. Test: TC-TMS-009	Depends on vehicle priority flag.

4.4 Traffic Monitoring and Statistics

Description: Monitor traffic density, queue sizes, and simulation statistics.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMS-F-010	The system shall calculate and display traffic density for each lane based on the number of waiting vehicles.	Functional	Medium	System User	AC-TMS-F-010: The system reports Low, Medium, or High density according to the queue size. Test: TC-TMS-010	Density is derived from current queue size.
TMS-F-011	The system shall display the current number of vehicles waiting in each lane.	Functional	Medium	System User	AC-TMS-F-011: Queue size displayed for each lane matches the number of vehicles stored in that lane. Test: TC-TMS-011	Depends on queue state.
TMS-F-012	The system shall maintain and display the total number of vehicles added and processed during the simulation.	Functional	Medium	System User	AC-TMS-F-012: Statistics increase when vehicles are added or processed and are displayed correctly. Test: TC-TMS-012	Counters reset when the simulation is reset.

4.5 Simulation Control and Validation

Description: Provide simulation reset, safe input handling, and controlled application exit

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TMS-F-013	The system shall allow the user to reset the simulation and clear all lane queues and statistics.	Functional	High	System User	AC-TMS-F-013: After reset, all queues are empty and vehicle counters return to their initial state. Test: TC-TMS-013	Reset affects the current simulation state only.
TMS-F-014	The system shall validate menu, lane, vehicle ID, and queue-related inputs and shall reject invalid values.	Functional	High	System User	AC-TMS-F-014: Invalid input is rejected and the user receives an appropriate error message without terminating the application unexpectedly. Test: TC-TMS-014	Input validation is required for safe operation.
TMS-F-015	The system shall provide an option for the user to exit the simulation and terminate the application normally.	Functional	Medium	System User	AC-TMS-F-015: Selecting Exit terminates the simulation without errors. Test: TC-TMS-015	No external resources need to be released.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow TMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TMS-NF-001	The system shall respond to normal menu operations within 2 seconds on a standard development computer.	Performance	High	At least 95% of 30 measured operations complete within 2 seconds. Test: TC-NF-001
TMS-NF-002	The system shall maintain correct queue ordering and counters throughout normal simulation use.	Reliability	High	100 consecutive add/process operations produce no queue corruption or

				counter mismatch. Test: TC-NF-002
TMS-NF-003	The system shall handle invalid user input without crashing or accessing data outside defined queue bounds.	Security / Robustness	High	Invalid menu, lane, and boundary inputs are rejected safely. Test: TC-NF-003
TMS-NF-004	The system shall provide clear and understandable command-line messages for all supported operations and errors.	Usability	Medium	User evaluation confirms all menu options and error states are understandable. Test: TC-NF-004
TMS-NF-005	The system shall be maintainable through modular functions for queue operations, vehicle processing, monitoring, statistics, and simulation control.	Maintainability	Medium	Static analysis passes and each major function can be tested independently. Test: TC-NF-005

5.1. Security

5.1.1 Security Objectives

TMS-SO-001: Preserve the integrity of simulation and queue data by preventing invalid inputs and out-of-bounds queue operations.

TMS-SO-002: Ensure predictable and safe application behavior by rejecting malformed or unsupported user input without exposing sensitive system information.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
TMS-SR-001	The system shall validate lane numbers before accessing a lane queue.	Security	High	AC-TMS-SR-001: Out-of-range lane values are rejected. Test: TC-SEC-001
TMS-SR-002	The system shall enforce the maximum queue capacity before inserting a vehicle.	Security	High	AC-TMS-SR-002: Insertion is refused when the queue is full and existing data remains unchanged. Test: TC-SEC-002
TMS-SR-003	The system shall validate vehicle IDs and reject invalid or unsupported values.	Security	Medium	AC-TMS-SR-003: Invalid vehicle IDs do not enter the queue. Test: TC-SEC-003

TMS-SR-004	The system shall not execute operating-system commands or accept executable input from the user.	Security	High	AC-TMS-SR-004: User input is treated only as simulation data and no system command is executed. Test: TC-SEC-004
TMS-SR-005	The system shall avoid storing or displaying confidential personal information because the simulation does not require such information.	Security	Medium	AC-TMS-SR-005: Output and stored vehicle data contain only simulation fields such as ID and priority. Test: TC-SEC-005

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements are implemented and verified, all five NFRs meet their stated measurements, no critical security requirement fails, and the RTM has corresponding test cases.

- Acceptance test suites: Vehicle Management, Multiple Lane Management, Traffic Signal and Priority, Traffic Monitoring and Statistics, Simulation Control, Performance, Robustness, Usability, and Security tests.

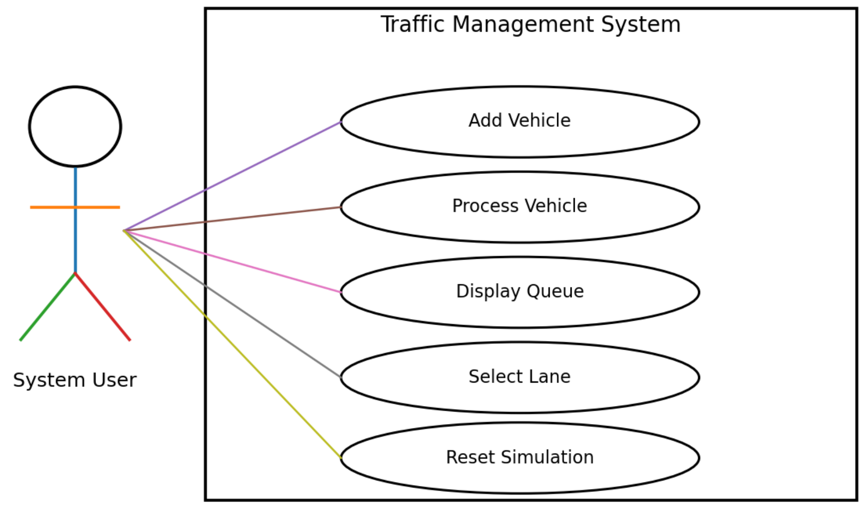
- Quality attributes evaluated: performance, reliability, security/robustness, usability, and maintainability.

7. System models and diagrams

7.1 UML Use-Case Diagram 1 – Vehicle and Lane Management

Actors and use cases for vehicle and lane operations are shown below.

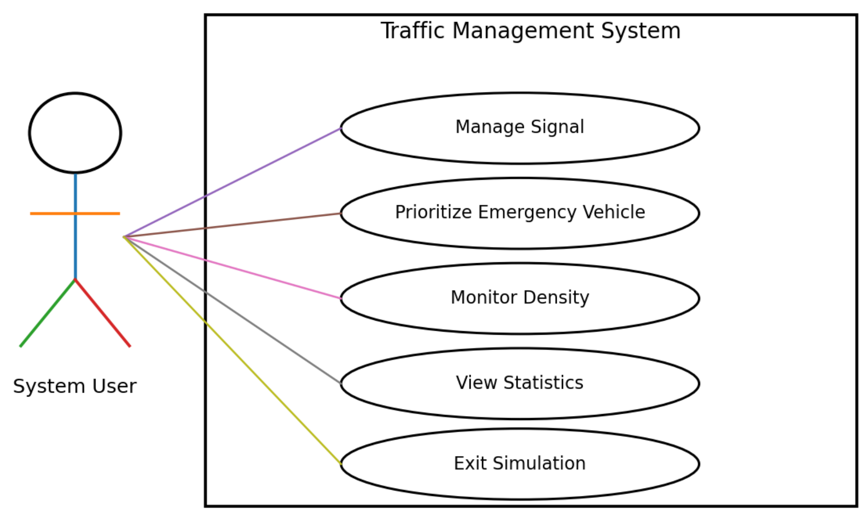
UML Use-Case Diagram 1



7.2 UML Use-Case Diagram 2 – Monitoring and Control

Actors and use cases for monitoring, signal, priority, and statistics are shown below.

UML Use-Case Diagram 2



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8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TMS-F-001	Add vehicle	4.1 / DS-TMS-001	VehicleModule	TC-TMS-001	N	Planned / to be verified
TMS-F-002	Process vehicle	4.1 / DS-TMS-002	VehicleModule	TC-TMS-002	N	Planned / to be verified
TMS-F-003	Display queue	4.1 / DS-TMS-003	DisplayModule	TC-TMS-003	N	Planned / to be verified
TMS-F-004	Multiple lanes	4.2 / DS-TMS-004	LaneModule	TC-TMS-004	N	Planned / to be verified
TMS-F-005	Lane selection	4.2 / DS-TMS-005	ValidationModule	TC-TMS-005	N	Planned / to be verified
TMS-F-006	Queue boundary handling	4.2 / DS-TMS-006	QueueModule	TC-TMS-006	N	Planned / to be verified
TMS-F-007	Traffic signal	4.3 / DS-TMS-007	SignalModule	TC-TMS-007	N	Planned / to be verified
TMS-F-008	Emergency priority	4.3 / DS-TMS-008	PriorityModule	TC-TMS-008	N	Planned / to be verified
TMS-F-009	Priority display	4.3 / DS-TMS-009	DisplayModule	TC-TMS-009	N	Planned / to be verified
TMS-F-010	Traffic density	4.4 / DS-TMS-010	MonitoringModule	TC-TMS-010	N	Planned / to be verified
TMS-F-011	Queue monitoring	4.4 / DS-TMS-011	MonitoringModule	TC-TMS-011	N	Planned / to be verified
TMS-F-012	Statistics	4.4 / DS-TMS-012	StatisticsModule	TC-TMS-012	N	Planned / to be verified
TMS-F-013	Simulation reset	4.5 / DS-TMS-013	SimulationModule	TC-TMS-013	N	Planned / to be verified
TMS-F-014	Input validation	4.5 / DS-TMS-014	ValidationModule	TC-TMS-014	N	Planned / to be verified
TMS-F-015	Exit simulation	4.5 / DS-TMS-015	SimulationModule	TC-TMS-015	N	Planned / to be verified
TMS-NF-001	Response time	5 / DS-NF-001	QualityModule	TC-NF-001	N	Planned / to be verified
TMS-NF-002	Queue reliability	5 / DS-NF-002	QualityModule	TC-NF-002	N	Planned / to be verified
TMS-NF-003	Input robustness	5 / DS-NF-003	QualityModule	TC-NF-003	N	Planned / to be verified
TMS-NF-004	CLI usability	5 / DS-NF-004	QualityModule	TC-NF-004	N	Planned / to be verified
TMS-NF-005	Maintainability	5 / DS-NF-005	QualityModule	TC-NF-005	N	Planned / to be verified
TMS-SR-001	Lane validation	5.1.2 / DS-SEC-001	SecurityModule	TC-SEC-001	N	Planned / to be verified
TMS-SR-002	Queue capacity	5.1.2 / DS-SEC-002	SecurityModule	TC-SEC-002	N	Planned / to be verified
TMS-SR-003	Vehicle ID validation	5.1.2 / DS-SEC-003	SecurityModule	TC-SEC-003	N	Planned / to be verified
TMS-SR-004	No command execution	5.1.2 / DS-SEC-004	SecurityModule	TC-SEC-004	N	Planned / to be verified
TMS-SR-005	No confidential data	5.1.2 / DS-SEC-005	SecurityModule	TC-SEC-005	N	Planned / to be verified